

Smart Accident Detection and Emergency Response ASIC

PROBLEM

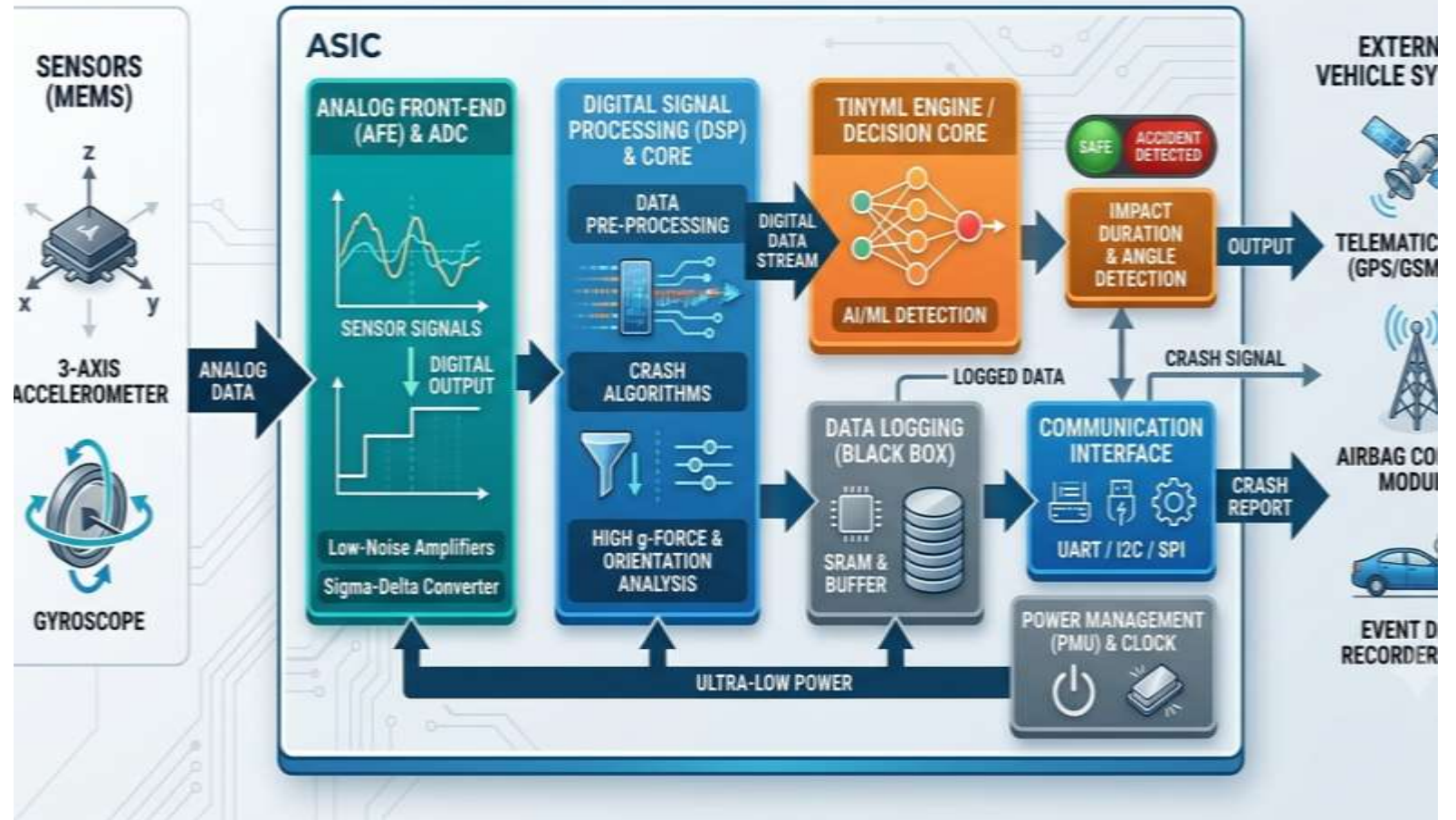
In India Road accidents are a major social challenge, leading to injuries and death

PROBLEM SOLUTION

- The Proposed **Smart Accident Detection ASIC** is a low-cost, energy-efficient chip designed to improve road safety in India. It continuously monitors vehicle movement using accelerometer, gyroscope, and vibration sensors to detect severe accidents in real time. When an accident is identified, the ASIC automatically retrieves the vehicle's location through GPS and sends emergency alerts to nearby hospitals, police stations, and family members using a wireless communication module.

BLOCK DIAGRAM

INTEGRATED BLOCK DIAGRAM: SMART ACCIDENT DETECTION ASIC (APPLICATION-SPECIFIC INTEGRATED CIRCUIT)



INPUTS,OUTPUT AND FUNCTIONAL BLOCK

- **Inputs**

- Acceleration data from accelerometer
- Angular motion data from gyroscope
- Impact/vibration sensor signals

- **Major Functional Blocks**

- Sensor Interface Circuits
- Analog Front End (AFE) and ADC
- Accident Detection Logic Unit

- **Outputs**

- Automatic accident detection signal
- Emergency alert messages with location details